

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 04/22/2020

Quiz 11: Section 4.7, 4.9

Name (Print): _____

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Section 4.7

3 pts

1. Evaluate the limit $\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2}$.

(A) 0

(B) $\frac{0}{0}$

(C) $\frac{1}{2}$

(D) The limit does not exist

Substituting $x = 0$ into this function produces the indeterminate form $\frac{0}{0}$. Applying L'Hôpital's Rule, we have

$$\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x},$$

which is another limit of the form $\frac{0}{0}$. Therefore, we apply L'Hôpital's Rule again:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}.$$

Hence, we may conclude that

$$\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2} = \frac{1}{2}.$$

It is easy to overlook a crucial step in this computation: After applying L'Hôpital's Rule the first time, you **must** establish that the new limit is an indeterminate form before applying L'Hôpital's Rule a second time.

3 pts

2. Evaluate the limit $\lim_{x \rightarrow 0^+} x^{x^2}$.Ⓐ 0^0

Ⓑ 1

Ⓒ 0

Ⓓ The limit does not exist

This limit has the form 0^0 . Using the given two-step procedure, we note that $x^{x^2} = e^{x^2 \ln(x)}$ and first evaluate $L = \lim_{x \rightarrow 0^+} x^2 \ln(x)$. This limit has the form $0 \cdot \infty$, which may be put in the form ∞/∞ so that L'Hôpital's Rule can be applied:

$$L = \lim_{x \rightarrow 0^+} x^2 \ln(x) = \lim_{x \rightarrow 0^+} \frac{\ln(x)}{\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} \frac{1/x}{-2/x^3} = \lim_{x \rightarrow 0^+} -\frac{1}{2}x^2 = 0.$$

Now, the second step is to exponentiate the limit:

$$\lim_{x \rightarrow 0^+} x^x = e^L = e^0 = 1.$$

Section 4.9

4 pts

1. Determine the indefinite integral $\int \left(\frac{4x^{19} - 5x^{-8}}{x^6} \right) dx$

Ⓐ $\frac{2}{7}x^{14} + \frac{5}{13}x^{-13} + C$

Ⓑ $\frac{(1/5)x^{20} + (5/9)x^{-9}}{(1/7)x^7} + C$

Ⓒ $\left(\frac{1}{7}x^{-7} \right) \left(\frac{1}{5}x^{20} + \frac{5}{7}x^{-7} \right) + C$

Ⓓ $\frac{2}{7}x^{14} + \frac{5}{14}x^{-15} + C$

$$\begin{aligned} \int \left(\frac{4x^{19} - 5x^{-8}}{x^6} \right) dx &= \int (4x^{13} - 5x^{-14}) dx \\ &= \int 4x^{13} dx - \int 5x^{-14} dx \\ &= 4 \int x^{13} dx - 5 \int x^{-14} dx \\ &= 4 \left(\frac{1}{14}x^{14} \right) - 5 \left(\frac{-1}{13}x^{-13} \right) + C \\ &= \frac{2}{7}x^{14} + \frac{5}{13}x^{-13} + C \end{aligned}$$